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ANSIBLE:

Simply It is a IT automation tool
its a Configuration Management Tool.

Launch Amazon Linux 2 , t2.micro

Installation, Ansible needs python

amazon-linux-extras install ansible2 -y

yum install python3 python-pip python-dlevel -y

ansible --version

Examples: Adhoc Commands

ansible -m ping localhost [Ping localhost]

ansible localhost -a "yum install git -y"

ansible localhost -a "git --version"

ansible localhost -a "yum install maven -y"

ansible localhost -a "mvn --version"

ansible localhost -a "touch file1"

ansible localhost -a "ls"

ansible localhost -a "yum install httpd -y"

ansible localhost -a "systemctl status httpd"

ansible localhost -a "systemctl start httpd"

ansible localhost -a "user add reyaz"

ansible localhost -a "cat /etc/passwd"

ansible localhost -a "yum remove git* maven* httpd* -y"

SETUP:

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CREATE 5 SERVERS - Amazon Linux 2 [1=ANSIBLE Master, 2=PROD, 2=DEV]

EXECUTE THE BELOW COMMANDS ON ALL SERVERS:

sudo -i

hostnamectl set-hostname ansible/prod-1/prod-2/dev-1/dev-2

sudo -i

Now Root user from ansible server needs to login to all servers using root
username and password

Do below commands in all servers using multi-exec

first set the password for root

passwd root

set new password: root123456

enable all server to login as root

```
vi /etc/ssh/sshd_config (38 & 61 uncomment both lines)
systemctl restart sshd
systemctl status sshd
hostname -i -- to see the private ip
```

Go to Ansible Master

Lets Generate SSH Keys, using this KEY Ansible server will communicate with worker nodes

```
ssh-keygen
```

```
ll .ssh
```

```
ssh-copy-id root@private ip of prod-1 -- > yes -- > password -- > ssh private ip
-- > ctrl d
ssh-copy-id root@private ip of prod-2 -- > yes -- > password -- > ssh private ip
-- > ctrl d
ssh-copy-id root@private ip of dev-1 -- > yes -- > password -- > ssh private ip
-- > ctrl d
ssh-copy-id root@private ip of dev-2 -- > yes -- > password -- > ssh private ip
-- > ctrl d
```

just to see the connection

```
ssh root@privateip
```

```
exit
```

Inventory file

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There are 2 types of inventory file, static and Dynamic

```
vi /etc/ansible/hosts
# Ex 1: Ungrouped hosts, specify before any group headers.
[prod]
172.31.20.40
172.31.21.25
[dev]
172.31.31.77
172.31.22.114
```

ansible-inventory --list --> This will show the inventory list

ansible -m ping all : To check worker node connection with ansible server.

1. ADHOC COMMANDS:

these are simple Linux commands.

ADHOC commands are great for tasks you repeat daily

These are used for temp works.

-a = argument

```
ansible all -a "yum update -y"
ansible all -a "yum install git -y"
ansible all -a "git --version"
ansible all -a "yum install maven -y"
ansible all -a "mvn --version"
```

```
ansible all -a "touch file1"
ansible all -a "touch reyaz.txt"
ansible all -a "ls"
ansible all -a "yum install httpd -y"
ansible all -a "systemctl status httpd"
ansible all -a "systemctl start httpd"
ansible all -a "user add reyaz"
ansible all -a "cat /etc/passwd"
ansible all -a "yum remove git* maven* httpd* -y"
```